Silent Defects in a Self-Coordinating Agent Fleet: A Taxonomy, a Peer-Review Record, and Three Withdrawn Claims

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***Abstract*—Conductor is a fleet of persistent, memory-divergent agent sessions — one base model, durable per-session context, a shared append-only bus, and a human gating only irreversible actions. Over a months-long live deployment the sessions co-designed and hardened their own coordination substrate in vivo under adversarial peer review, closing roughly twenty named failure modes, and caught one another's defects from divergent empirical vantages. We report two things the deployment supports. The first is a set of recurring shapes of *silent* failure in agent self-evaluation — failures that return a plausible answer rather than an error, and so survive every gate that checks for errors. Each names a place a verification must go; one is specific to agents carrying accumulated context, and we believe it is new. We do not claim the set is reusable beyond this deployment, and we report a merge a reader should try first. The second is that peer sessions catch defects their author and the human miss, evidenced from the timestamped record rather than recalled. We also report, at length, what the deployment could *not* establish. We set out to test whether accumulated context makes a session more capable on task $N{+}1$; an earlier version of this paper reported that it does not, and we withdraw that negative. The ablation behind it never manipulated the construct it named, and no arm of any *executable* task ever failed, so its falsifier was unreachable. The pre-registered replacement cannot be fielded here at all: a fleet running one session per repository has no second session with the same history, so it cannot furnish a within-domain control arm for any claim about accumulated context — a property of the architecture, not of the hypothesis. Two further claims are withdrawn on defects found by outside review after we considered the paper ready, both in the direction that flattered it, and we report them as specimens of our own taxonomy: a longitudinal human-attention trend whose instrument drops records at a rate that rises with fleet size — the same direction as the claimed effect — and a pre-registered baseline whose target turned out to be one day old, measuring authorship recency rather than accumulated context. The practitioner consequences we still stand behind: commit everything that must stay true to the carrier any fresh session can read; give silent-failure classes named remedies rather than a general injunction to be careful; and treat a peer with a different vantage as an instrument, not a courtesy.**

***Index Terms*—LLM agents, multi-agent systems, agent memory, coordination substrate, human-in-the-loop, experience report, ablation study, self-adaptive systems**

I. Introduction and Contributions

THE prevailing agentic pattern is an orchestrator dispatching to stateless workers — reproducible, but *permanently at task 1*: no carried expertise, no division of labour, no shared context. Conductor inverts the premise: long-lived sessions with divergent memories talk to each other and build their own coordination substrate.

This paper reports what that deployment established, and it is narrower than what we set out to show. Two contributions survive contact with their own evidence:

1) **Recurring shapes of silent failure in agent self-evaluation** (§V-A), surfaced by writing and ablating the case corpus — *not*, despite an earlier version of this claim, by classifying the $\sim$20 substrate failure modes we closed; we never built that mapping, and most of those fixes are mechanism bugs this classification does not fit. Several classes are ablation-structured — disable the mechanism and the failure returns — and at least one ablated in operation. One class (thoroughness regression) is specific to agents carrying accumulated context, and we believe it is new. What the classification buys is *where to point a verification*, not a separate discipline per class: the remedies collapse toward this paper's own through-line, verify the outcome rather than the intention (§III), and we say so rather than claim an irreducibility we cannot demonstrate.
2) **Cross-context peer review catches defects that individual sessions and the human miss** (§V-C), evidenced from the timestamped record: A ships at t_0, B refutes at t_1, both attested. This is the claim the deployment supports best *on its own record* — while being, per our own tier rule, the least reviewer-checkable, since the underlying bus log is private and only a curated extract ships. Both things are true and the second bounds the first. It is the one we were least expecting to lead with.

The system and the method that produced them — in-vivo adversarial co-design under reversibility-calibrated HITL, where primitives were *provoked by silent-loss failures* rather than designed up front, and a human's approval is scoped to *irreversibility* (commits are free; pushes, and anything outliving the session, are gated) — are reported as the *setting*, not as a validated result. We never ran the method against an engineer specifying the primitives up front, so we claim only that it is replicable, has a stated mechanism, and demonstrably produced a working substrate.

This is an experience report and we ask to be judged at that bar: the contribution is what a months-long real deployment

reveals. We claim neither statistical generality (N=1 deployment) nor emergent intelligence.

What we cannot claim, stated first because an earlier draft claimed it. We set out to show that capability compounds — that a context-carrying session solves what a fresh one cannot. It does not appear here, for two separate reasons, and neither is a null result. Our original ablation never manipulated the construct it named, and no arm of any *executable* task ever failed, so the falsifier was unreachable (§V-D). The pre-registered replacement then could not be fielded at all: this fleet runs one session per repository, so the only domain able to supply six live context-carrying arms is the one the authors wrote (§V-D). Per that pre-registration's committed interpretation, inability to construct a discriminating task is reported as a *measurement-capability limit of the deployment* and never as evidence for the negative.

A. What each contribution rests on, and what would falsify it

The two contributions are not epistemically equal, and neither is a controlled result.

The taxonomy (contribution 1) rests on failure modes we closed and can point at: the fix commits exist and several are ablation-structured; the classification itself, however, came from writing and ablating the cases, not from those commits. We do not claim the classes were derived rather than imposed: the same fleet, over an overlapping corpus, produced a different and non-isomorphic classification of its own silent failures, which is evidence that a classification here is author-contingent. *Falsifier:* the taxonomy weakens if its classes collapse into one another under a larger corpus. We do not offer "a single generic discipline would have caught these" as a falsifier, because one plausibly would: our own through-line, verify the outcome rather than the intention, covers the remedy of every class here. The classification's claim is weaker and we state it as such — it names *where* the verification goes. We note the merge a reader should try first: classes 1, 3 and 4 all describe a provenance label outrunning its warrant, and we have no outcome-difference argument that keeps them apart. We report the classes we did not merge; a reader who can merge them has refuted this.

Peer catching (contribution 2) rests on the timestamped record: A ships at t_0, B refutes at t_1. *Falsifier:* it weakens if the catches would have been made as readily by ordinary single-author review, or if the peer's advantage is really access rather than vantage — B seeing a file A could not. We do *not* adjudicate access versus vantage row by row — we promised that in an earlier draft and never built it, and several rows are plainly access (a file on a shared disk, a donor snippet read in review). What §V-C does separate is whether a measurement vantage *exists* from whether a peer *used* it, because an earlier draft conflated them. *And a limit we cannot engineer away:* the fleet's primary record is an append-only message log held outside the public repository, so catches are attested by us and verifiable by a reader only through the curated extract we publish. The classification is checkable; the completeness of the underlying log is not.

TABLE I

HUMAN TOUCH PER ACTIVE PROJECT (`HISTORY.JSONL`). JAN/MAR (1 AND 7 PROMPTS) ARE OMITTED; APRIL, THE SUBSTRATE'S BUILD-OUT MONTH, IS SHOWN BUT EXCLUDED FROM THE SHAPE CLAIM AS BOOTSTRAP-HEAVY. THREE BUCKETS CANNOT ESTABLISH A TREND AND JUNE ≈ MAY, SO WE *reported* THE SHAPE. WE NO LONGER CLAIM IT: THIS TABLE IS RETAINED AS THE CLASS-2 SPECIMEN OF §V-A, BECAUSE ITS INSTRUMENT UNDER-COUNTS AT A RATE THAT RISES WITH CONCURRENCY — THE SAME DIRECTION AS THE EFFECT — AND THE TABLE IS INTERNALLY CONSISTENT ANYWAY, WHICH IS WHAT MAKES THE DEFECT INVISIBLE FROM INSIDE THE DATA. THE WITHDRAWN ARGUMENT, AND A 14-WEEK BREAKDOWN, ARE PRESERVED IN THE REPLICATION PACKAGE (AS-OF 2026-07-27 — FOUR DAYS BEHIND THIS TABLE'S 2026-07-31 STAMP, SO ITS JULY ROW READS 1,219/34/35.9; THE DIFFERENCE IS FOUR DAYS OF JULY). ANALYZED IN §V-B.

Month	Human prompts	Active projects	Prompts / project
2026-04	2,027	10	202.7
2026-05	1,117	16	69.8
2026-06	1,936	27	71.7
2026-07	1,270	35	36.3

The method has no controlled evidence at all and we do not claim it is superior to up-front design. *Falsifier:* the method claim weakens if the primitives it converged on are worse than a competent engineer's up-front design — a comparison we did not run.

A note on the direction of our own errors. Every correction forced on this paper by review moved a claim *down*: a negative we called solid was unsound, a signal we called supporting was confounded, and two catch attributions were wrong in the flattering direction. We have found no instance of review moving one of our claims up. A reader should weight the remaining claims accordingly, and we say so here rather than in a closing threats section, where it would arrive too late to be used.

II. RELATED WORK

(Full treatment: `related-work.md`.) Capability-aware routing exists but is keyed on declared role (AutoGen [1]; MetaGPT [2]; ChatDev [3]; CrewAI, *stateless by default*) or predicted fit (LangGraph supervisor; DyLAN [4]; CaptainAgent [5]; AutoAgents [6]; MasRouter [7]) over agents that are stateless-by-default — with query-level model routers (RouteLLM [8], FrugalGPT [9]) a separate axis, and Mixture-of-Agents [10] an ensemble that aggregates all proposers, not a router (reclassified per review). Voyager [11] accumulates competence for a *single* agent; ADAS [12] / Darwin Gödel Machine [13] evolve a *population of agent designs* at design-search time; MAPE-K (Kephart & Chess, *Computer* 36(1), 2003) [14] is the classical self-adaptive-control analogue.

Prior art the substrate rediscovers (and we name it as such). Several of Conductor's primitives are not new in isolation — and saying so is evidence *for* the method, not against it: an adversarial process that independently reconverged on known load-bearing designs is weak-but-real evidence the process finds *real* designs. The append-only shared log that peers read and write, with a nominated lead coordinating, is a blackboard architecture (Hearsay-II [15]; the blackboard-

for-control model [16]); the verified-order lifecycle (deliver only if the artifact landed) is two-phase commit; the human approval bound to an action token rather than conveyed in prose is capability-based authorization; and the design loop itself rediscovers evolutionary prototyping [17], [18], the spiral model [19], and game-day/chaos-engineering practice [20]. Our contribution is not these mechanisms but the *setting* — long-lived, context-heavy, same-model peers rediscovering and composing them in vivo under a reversibility-gating human.

Honest positioning (we do not overclaim novelty). A 2025 memory-augmented-MAS line does pursue cross-task team competence — most pointedly G-Memory [21] (hierarchical memory "nurturing the progressive evolution of agent teams," i.e. task $N{+}1$ better because of $1 \ldots N$ across a team), plus RCR-Router [22] (routing *structured memory to agents within a task*) and the transactive-memory framing for agents. So "team competence accumulates" is *not* our novelty. Two things differ. (i) Each peer is a long-lived, human-in-the-loop *session identity* that lived the work — not an ephemeral agent doing retrieval over a shared memory store. We are careful *not* to state what that buys: whether a lived session identity raises outcome capability over a fresh session holding the same committed carrier is exactly what this deployment could not measure (§V-D), and an earlier version of this paper wrongly reported that it does not. What we do claim here is narrower and does not depend on that question: a lead can route work by a peer's demonstrated history over real artifacts rather than by a declared persona. (ii) Routing is therefore a lead's grounded judgment over real artifacts (bus, asset cards, `CLAUDE.md`), keyed on lived history rather than a self-described role. We do *not* list "experience report vs. benchmark evaluation" as a third differentiator: that is a difference in *rigour*, and in the direction of less. We frame our contribution as *under-explored relative to* that memory-augmented line, from which we differ in kind (efficiency-and-routing from *lived* identities, studied as a running deployment) — not as a gap nobody has touched.

III. The Architecture

One base model; each session a persistent identity with durable memory; a read-only observer (Conductor) that never drives the agents (Fig. 1). Primitives, each landed as a versioned mechanism in operation:

- Bus — append-only shared log; directed mail is auto-delivered (idle recipient woken), which is the courier-elimination mechanism (and an *unreliable* one — §V-B).
- Verified orders — `deliver` reads the artifact back and refuses unless it landed.
- Project layer — a nominated lead drafts a job-DAG plan the human approves at one gate, then fans jobs (directed orders) along dependency edges; a *peer*, not the lead, accepts the lead's own deliverables.
- Decision shield — worker decides technical calls and logs them; lead decides project calls; a denylist + severity hatch go straight to the human, and the lead is *structurally barred* from deciding them.

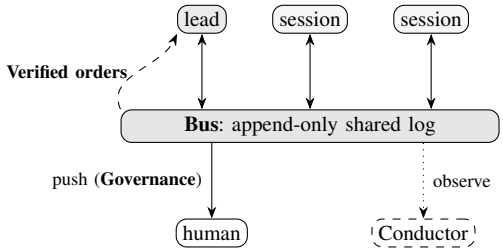


Fig. 1. The composition: persistent *sessions* post to and read an append-only **Bus**; a lead fans work as **Verified orders** (`deliver` reads the artifact back before acknowledging); the *human* gates only irreversible acts (**Governance**: the SHA-pinned push gate); *Conductor* observes read-only and never drives the agents. The through-line: *verify the outcome, not the intention.*

- Governance — SHA-pinned push gate; observer-owned admission throttle (not the lead — fox/henhouse); a measured per-project token meter (estimation is theater; only the live meter gates).

The through-line is *verify the outcome, not the intention*.

IV. The Method: In-Vivo Adversarial Co-Design

The design loop that produced the above: (1) a session hits a *silent-loss* failure in real work (a tool reports success while something is quietly wrong); (2) it publishes the specimen to the bus; (3) context-divergent peers attack the proposed fix from different empirical vantages (their histories genuinely differ); (4) the human gates only the irreversible. Their independence is only *relative to* that divergence — not one model echoing itself, but not uncorrelated errors either; bounded in §V-C.

The loop's *shape* is deliberately unoriginal and we claim no novelty for it: operating a system to learn what it should have required is evolutionary prototyping [17], [18], the spiral model [19], and game-day/chaos-engineering practice [20]. One post-corpus exercise (2026-08-05, outside the stamp) cuts against the usual prototyping argument: the first live run of a newly-shipped fleet wind-down protocol, across 31 sessions, surfaced two silent-success root causes, both in taxonomy class 2 (§V-A) and both violating the architecture's own through-line, *verify the outcome, not the intention* (§III). The instructive part is not that the drill found them but that **the requirement they violate was already published in our own record months earlier** and simply not enforced on a later actuator. The gap was **compliance, not knowledge** — weaker and more useful than requirements-unknowability: a through-line stated once does not propagate itself to each new actuator, and the drill is what detects the drift.

V. Evaluation

A note on names: throughout the evaluation, a codename in parentheses (e.g. `net-emu`, `perf-A`) identifies an *individual Claude Code session* — the participant that produced, or is the subject of, the adjacent claim. These codenames are role-hinting pseudonyms for publication. We state the limit of that layer plainly rather than implying more: the artifact files carry the deployment's own session names, so the pseudonymisation

TABLE II
PROVENANCE TIERS USED THROUGHOUT THE EVALUATION.

Tier	Meaning
MEASURED	ran it on a censused system, with a resolvable receipt (commit/log)
RECALLED	from memory/narration; may be wrong, flagged as such
GAP	the honest absence of a measurement we would want
LANDED	a mechanism shipped and is live in the running system
NULL	a pre-registered test whose falsifier fired (reported as such)
reproducible-in-principle	a recipe that outlives the record but has not been re-executed

is a readability device in the prose and not a confidentiality measure, and the correspondence is recoverable from the filenames cited beside each claim. Every quantitative claim below carries an explicit *provenance tier*; the fleet adopted these as a discipline and we define them once here (Table II). The rule the deployment enforces on itself: a derived or recalled number may never be reported bare, and a claim a reviewer cannot resolve is, at that moment, not yet evidence.

Corpus, MEASURED from git + bus + `history.jsonl`, as-of 2026-07-31 (every count carries this stamp — the log is append-only and non-stationary, so a bare number is under-specified; events after the stamp are reported as dated addenda and are *never* counted, in either direction): 290 commits (38 of type `fix`); 52 version landings; 2,744 bus messages, of which 2,739 are non-system, from 56 senders (1,109 directed + 1,107 broadcast + 521 announcement + 2 unclassified = 2,739; directed share 1,109/2,739 = 40.5%); 6,358 human prompts across 47 projects back to 2026-01-14. **Division of labour is spread but with a real head, not a power law** — the concentration computed by the harvester (`evidence-harvest.py _concentration`), the addressing classified from the same log, over *all* non-system senders: Gini 0.698, top sender 9.4%, top-3 25.4%, and 13 of 56 senders to reach 80%. Many peers carry load, no single dominant author, but a genuine concentration we report rather than round off to "distributed." Reflexivity confound — controlled, not just conceded. Because 196 of the 2,739 non-system messages are this paper's own review/ablation traffic, a concentration statistic over the whole log could be an artifact of the act of measuring. We therefore recompute it *restricting to* messages before the paper effort began (2026-07-24; n=2,543 non-system messages, 52 senders): Gini 0.693, top sender 9.4%, top-3 26.4%, 13 of 52 senders to 80% — *indistinguishable* from the full-log figures above (Gini 0.693 vs. 0.698; top-3 26.4% vs. 25.4%). The head is a property of the *organic* deployment, not of the paper's own threads; the reflexivity we flag elsewhere does *not* contaminate this result (both numbers are shown). Why roles emerged at all — no one assigned them. Specialization was not designed; it is an attractor of two forces the substrate creates. Continuity and ownership: a session that built a subsystem holds the most context about it, so the cheapest place to route the next related task is that same session — and routing it there deepens the advantage, a positive feedback that concentrates each domain onto whoever first worked it. Repeated exposure then hardens it: memory and the bus make yesterday's work legible today, so a returning session resumes its domain rather than re-learning a new one. That the resulting head is a *spread* one (Gini 0.698, no dominant author) is because these forces are *local* — one attractor per domain — not because anyone balanced the load. Role identity is an emergent property of continuity-plus-routing, not an org chart — which is why it is the foundation the later findings rest on rather than a policy that could be tuned away.

A. Failure modes closed, and the taxonomy the case corpus surfaced (RQ2)

MEASURED: of the 38 `fix` commits (as-of the corpus stamp), ∼20 close *named coordination failure modes*. Several are ablation-structured — disable the mechanism, the failure returns — and at least one ablation *ran in operation*: the SHA-pinned push token, unpinned, would let an approval for commit A push commit B; the gate refused a stale approval live during this project. The two-phase-commit and member-cursor ablations are specified and pending.

A defect taxonomy the fleet converged on (a secondary contribution). Writing and ablating the cases surfaced a small classification of the *silent* defects the method exists to catch. Each names a place a verification must go; the verifications themselves are variations on executing the thing you currently believe unexecuted:

1) Single sample promoted to a property — a condition true of one run asserted as general. Remedy: replicate + tag. Sharpened (perf-A): a *measured* number promoted past its conditions is *worse* than a derived one, because the `measured` badge actively defends the error.
2) Green-but-wrong / wired-but-broken in the untested direction — a gate green + documented + shipped, passing every gate-ON test while the gate-OFF path leaks (emu-C, net-emu). Replaying the green measurement *n* times passes *n* times; the only fix is asserting a different measurement — execute the off-state / your own reproduction once. A post-corpus specimen adds an axis: the untested direction can be **selected by the very condition the mechanism exists to attest**. The acknowledgment routine of §V-C's addendum reached its fatal path only when a session had *exactly zero* unpushed commits, so failure concentrated on sessions satisfying the very condition the ack certifies — **informative missingness** (MNAR, [23]) in a monitoring mechanism, so the census is *biased*, not merely thinned. Here the class-2 remedy is necessary but insufficient: **zero unpushed is not the off-state; it is the attested value, on the nominal success path** (the off-state, a dirty tree, *was* exercised), so the remedy extends to the boundary value of the predicate the mechanism selects on.

3) Provenance-tier silently dropped during remediation — re-anchoring a number to a *recipe* downgrades MEASURED → reproducible-in-principle until someone runs it (band, emu-C). Remedy: run the recipe and *say you did* (emu-C did — the mutation reproduces `1,409,307,648`).
4) Conservative-error-is-durable — a wrong-but-safe-looking value evades the provenance check that would normally catch it. "A conservative error is not a safe error; it is a durable one." Where the verification goes: the provenance check must not exempt a value for looking safe — a conservative figure needs its warrant established on the same terms as a flattering one, and that is the step the safe-looking direction reliably skips.
5) Thoroughness regression — the context shortcut that made the agent fast made it skip the secondary gap (mcu-emu DISMAP; perf-B's third site). Remedy: re-derive under a differently-drawn boundary / forced first-principles trace.

And the remedy split the same convergence produced: TAG (cheap disclosure; catches a condition you measured but did not state) → ABLATE (an invariant/control; catches a condition you *believed held and did not*) → HARVEST (a parallel peer at the same boundary building the same artifact is a *naturally-occurring* off-state ablation — emu-C↔emu-B, perf-A↔perf-B — cheaper than a designed one, and record-visible: cross-tree agreement on different fixes is derived-not-copied). *Shipping only the tag documents the bug rather than fixing it.*

B. Specimen 1 — courier elimination: a withdrawn claim (RQ1)

We do not claim this result. We report it because the way it failed is a specimen of class 2 in §V-A, caught in this paper's own headline measurement. The argument as we originally made it — a monthly per-project human-touch decline, a weekly series read for direction, and a leverage computation over concurrent projects — is preserved in the replication package; we do not restate it here, because every figure in it derives from the instrument the next paragraph invalidates. Table I is retained as the exhibit.

The defect, and why it is fatal rather than a caveat. The instrument is `history.jsonl`, and it loses records. The writer truncates its own append under contention, so the loss is *concentrated rather than random, and it scales with fleet size*. That is the whole problem: the claimed effect is a decline in per-project human touch as concurrent projects rise, and the instrument drops more records exactly as concurrency rises. A decline would therefore appear in this measurement whether or not one occurred. The direction of the artifact and the direction of the claim are the same, so no amount of internal consistency in the table can separate them — and the table *is* internally consistent, which is precisely what made the defect invisible from inside the data.

It is class-2 in §V-A's terms: informative missingness (MNAR [23]). The failures of a monitoring mechanism concentrate on the very condition the mechanism exists to attest, so the census is *biased*, not merely thinned. We wrote that class from the acknowledgment-routine defect (§V-C addendum) and did not apply it to this section's own measurement. **Knowing the class did not protect the author who wrote it** — which we report as the only field trial of the taxonomy's prospective value in this paper, and it was negative.

What survives, explicitly. The sender decomposition does *not* ride this instrument. Categorizing every bus sender (MEASURED, `human_fleet_decomposition.md`) as human (`[operator]`) or fleet gives the human 0.9% of all coordination messages and 1.9% of directed messages. That figure rests on the bus log, which has a different writer and no contention-correlated loss, and we retain it. It speaks to who originates coordination — the sharpest objection to the method claim, that the human is the real router — and it stands independently of the withdrawn trend.

What we would need to claim it. An independent record of human prompts, written by a different mechanism than the one under test, and a completeness check against it — which is a second enumeration, not a replication of the first. We did not have one, and a replication of a biased instrument reproduces the bias.

C. RQ3 — Defect discovery by vantage (attested from the record; see the tier caveat)

We split two claims review conflated. (i) A measurement vantage exists (supporting, not the RQ3 claim): MEASURED (`cases_jaws.md`), a 38-Bash-call build had 17 measurement calls (1 `tracemalloc`, 12 `/proc` probes, 4 runs) — a vantage a browser assistant could not occupy, proven from the event log. (ii) A peer caught what the author shipped (the actual RQ3 claim, and it needs the *same* timestamp rigor — A ships at t0, B refutes at t1, both from the record). We enumerate the record-attested cross-session catches (13 attested, 12 classifiable — one further row, an ablation blind-arm outcome, is excluded by the corpus's own pre-registered rule against ablation-arm outcomes, and we removed it after review caught us counting it; tabulated in the replication package) and here name four representative ones, each with *attribution corrected* against the record after review flagged two mis-attributions in earlier drafts (a "`api-svc` tag-flip caught by a peer" was in fact self-caught, and a claimed "slam-A caught four errors" had no record — both removed). image-gen → api-svc: api-svc's RTX-5090 idle-power *denominator* (64.97 W) was contaminated by image-gen's own resident tenant; a clean board idles near 21 W, a physical floor image-gen held. doc-A → slam-A: slam-A's "fixed-K Amdahl overhead" bandwidth model, refuted by diffing two derivations on the shared tensor table (K doubles when the workload doubles, so a fixed overhead is impossible — the model refuted itself on its own arithmetic). media-npu → socdev-A: socdev-A's "verified" NPU INT8 model was quantized *without calibration normalization* (wrong input scale); media-npu reproduced its exact numbers to five decimals, then corrected the parameter. bench-A ↔ net-emu: bench-A's "+64 s rejoin stall" was its

own scorer's arrival-stamp backlog — net-emu relocated the measurement *into the guest* (a guest-emitted timestamp bench-A structurally cannot make; a pinned commit in the replication package) and a subsequent scorer replay put guest-side re-acquisition at 0.0 s. We keep the two artifacts distinct (the instrument commit vs. the replayed figure), pinned t0 (stall reported) $<$ t1 (refuted). A record-attested instance from the ablations: in `ablation_perf-A` a blind, zero-context arm caught a real overstatement in the author's own case (129 defective cells reachable via the engine API but *not* the shipped UI) in nine tool calls. **Cases are illustration; the events (order claims, timestamps, tool-call logs) are the evidence.**

Post-corpus addendum (2026-08-05, outside the as-of stamp; *not* counted in the 13/12 above). The catches above are peers refuting a peer's technical *result*; this one differs in kind — the defective artifact was the **coordination substrate itself** and the catching vantage was *being subject to it*, so it supports the §V-C bound rather than incrementing it. *Disclosure, load-bearing here:* `substrate-dev` below authored the wind-down machinery **and drafted this paper**; the catch is reported against ourselves. All five events are timestamped in the append-only record — **four on the bus, one in git** — on one day. **11:34:25**: `mcu-emu`, winding itself down, publishes the root cause of an acknowledgment routine that wrote no record — a counter that prints `0` and *exits non-zero*, terminating the routine under `set -e`. **14:32:54**: `substrate-dev` commits a fix for a *different* defect in the same routine. **17:17:25**: it broadcasts to the fleet that the bug is fixed. **17:20:53** (**3m28s later**): `band` measures the shipped routine from a clean tree — no output, exit 1, no record written. **18:09:05**: `band` publishes an audit enumerating **7 acknowledged against 24 not** (31 registered tags). Both commits are pinned in the replication package. We record it in **both** directions. As a vantage instance it is strong — the refutation came from *operating* the mechanism, timestamped against `substrate-dev`'s own claim. As a **substrate failure** it is equally real: the refutation sat on the bus **at least 5h43m** before the broadcast, so the delivery machinery this paper evaluates never put it in front of the session that authored the claim. **A vantage published but not *consumed* is a bound on the substrate, not only a credit to the peer.** The episode is a class-3 instance of our own taxonomy (§V-A), violating a remedy we had published; being **empirical-vantage** — a measured exit status, not a same-input reasoning disagreement — it leaves the bound below **unaffected**.

The bound on what these catches show. Every catch we count is *empirical*, and the load-bearing ones are *vantage*: the catcher held a different measurement position — it owned the contending tenant, held the silicon, ran the opposite test direction — and that position, not superior reasoning, produced the catch. **There is no record of a peer catching an author's reasoning error from the same inputs.** The sessions are therefore not independent estimators in any general sense; their independence is relative to the divergence in what they could measure, and nothing here supports a claim beyond that. Two consequences we hold to: before crediting two results as corroboration, check whether they share a generative mechanism and discount if they do; and a peer with the same vantage as the author is not a second instrument.

D. Specimen 2 — compounding capability: withdrawn, and why it could not be retested here (RQ4)

We set out to show that a session carrying lived context solves what a fresh one cannot. An earlier draft reported the negative — that it does not — and called it the paper's most solid evaluated result. **We withdraw both the claim and the negative.** The study could not detect the effect whose absence it reported, for two independent reasons, each verified against our own artifacts after an adversarial review raised them.

The ablation never manipulated the construct it named. The text described "a genuinely isolated fresh session versus a context-carrying one." The ablation record shows *both arms were fresh memoryless subagents*, differing only by a hand-distilled summary placed in the prompt. What was refuted, therefore, is that a summary injected into a prompt helps — a different and much weaker proposition. No arm of any ablation ever ran a live context-carrying session against a fresh one.

And the falsifier was unreachable. Across six ablations and the baseline, *no arm of any executable task ever failed* — and the one exception cuts the same way, since `emu-B`'s disposition-only runs saw *every* arm ship the wrong value, separating no arms either. A design whose falsifier requires the fresh arm to fail cannot fire on tasks nobody fails; the uniform success is a ceiling effect, not a null.

The replacement, and the limit that stopped it: We pre-registered a corrected experiment before running anything: fixed N, interleaved arms, isolation established by construction and transcript audit rather than self-report, a rubric hashed with the file, and grading by a session that authored neither the code under test nor the registration. It was witnessed by non-authors, amended three times pre-results in response to their objections, and every amendment is a pure append — the registered text is byte-identical under every later revision.

It has not run, and on this deployment it cannot. A qualifying task must have a live context-carrying arm with genuine domain history, at the registered N. **This fleet runs one session per repository**, so the only domain that can supply that many live arms is the one the authors wrote — which fails the registration's own author-circularity criterion. Three candidate tasks were nominated and disqualified for three structurally different reasons (access, cheap re-derivation, and author-circularity); the constraint is not candidate quality.

The registration committed in advance to how this outcome is read, precisely because it flatters a reframe we were separately considering: *inability to construct a discriminating task is reported as a measurement-capability limit of the deployment, and never as support for the negative.* We report it as that. The generalisable form is a property of the architecture rather than of the hypothesis: **a one-session-per-repository fleet cannot furnish a within-domain control arm for any**

claim about accumulated context, because the only session with the history is the only session there is.

E. Specimen 3 — the baseline: reported, not claimed (RQ5)

We pre-registered and ran one matched task both ways: a fresh memoryless session against the peer with the most lived history in the target file, both from an identical repo copy, graded on six behaviours by a black-box acceptance test, with the directional prediction committed before either arm ran and the instrument first proven satisfiable against an unmodified baseline.

The design stands; the target did not. Review established that the file chosen as "the peer's most lived history" was roughly one day old at baseline. Whatever the arms differed in, it was not accumulated context — it was authorship recency, which is a different variable with a different meaning. We therefore report the experiment and not its outcome.

We keep it here for two reasons. The apparatus is reusable and is pinned in the replication package — a later study with a correctly aged target can run it unchanged. And the failure is itself a member of §V-A's taxonomy: a pre-registered experiment, correctly executed, measuring a quantity other than the one its variable was named for. Pre-registration constrains *analysis*; it does not verify that the independent variable is the construct.

F. The paper as an instrument (an operational finding, not a controlled result)

Asking sessions to substantiate claims *to an external audience* forced verification of things taken on trust internally. In git: `perf-A` found a 46-day defect (129 cells rendering wrong fps, count never zero, never visible) while writing its case — after 46 days of use/test/review found nothing; `perf-B` surfaced a validation-across-an-unsafe-version provenance defect the same way. We state the hit-rate: of 19 case-writing sessions, 4 ($\sim$21%) reported a correction surfaced during the writing (`perf-A`, `perf-B`, `meter-A`, the RQ5 apparatus's own frozen-test defect); the rest reported none. We are careful about what this is and is not. It has no control arm: we did not measure how many defects surface in 19 sessions of *equivalent effort spent not writing for an external audience*, so 21% has nothing to beat, and the underlying effect — careful explanation to an outside reader surfaces latent defects — is well established for design documents and code review; *the only novel element here is that the writers were agents*. And it is not fully "testimony-free": the *commits* are machine evidence, but the *attribution* ("the defect was surfaced by the act of writing") is a causal judgment made and reported by the same sessions — two different epistemic objects, which we keep separate. So we report it as an **unanticipated operational finding with a stated hit-rate and no control arm** — worth recording, not a headline result.

VI. Implications for Practitioners

The measured results reduce to four working rules; each carries the bounds of §VII.

1. Fund the carrier, not the session. Effort that moves knowledge into the committed, executable carrier — code, asserts, invariants, checked-in operating docs — carries the work: every fresh arm in the executable-task ablations reached the fix, *on tasks of that kind*. Because every such arm succeeded, this is measured where capability was never the binding constraint — so it is evidence the carrier suffices, and *not* evidence about capability either way (§V-D). What effort spent preserving session memory buys instead is a question this deployment could not answer. When the two compete for budget, the carrier wins.

2. Route on demonstrated history — and audit the sessions you route to. A lead can route work by what a peer has actually done rather than by a declared role, and that is what the record supports. *We no longer quote an efficiency figure for it:* our matched-task numbers came from the baseline whose target proved a day old (§V-E), so they measure authorship recency, not accumulated context. But context also produced the one failure class unique to it: the *thoroughness regression*, in which a context-carrying session pattern-matched a staged answer and stopped short of the second defect a blind arm caught (§V-A, class 5; `ablation_mcu-emu.md`). Long-lived sessions therefore need a periodic fresh-eyes pass — a blind clone on the same task — as a standing control.

3. Write fixes into the executable carrier, not into prose. The stale-doc ablation found a wrong prose carrier to be "neither the asset nor the liability": every arm detected the staleness against the code (`ablation_perf-A.md`; the ablation is in the replication package, not restated here). Prose narrates; code, tests, and invariants govern. A fix that exists only as narration has not compounded.

4. Copy the cheap primitives first. The substrate's load-bearing mechanisms — verified orders that read the artifact back before acknowledging, SHA-pinned approval tokens, an observer-owned admission throttle — are small, model-agnostic, and follow one rule: *verify the outcome, not the intention* (§III). They transfer whether or not one adopts the fleet.

VII. Threats to Validity

- Self-evidencing circularity. A corpus of first-person cases about coordination, curated for a paper about coordination, is self-report at scale. Mitigation: cases are demoted to illustration; every load-bearing claim rests on the mechanical record.
- Single-model confound. Divergence is memory/context, not different minds; we own the composition and do not claim emergence.
- Selection bias / no denominator. The open call collects confirming specimens by construction. We hold $\geq$1 deliberate counter-case (a bug recognized instantly but re-tripped twice — recognition cost $\to$0, prevention cost unchanged), frame the corpus as illustrative, and note the missing denominator explicitly.
- 30-day record horizon. The platform silently deleted transcripts $>$30 days (now disabled); the pre-June

transcript record is GAP. Longitudinal claims use `history.jsonl` + git, which survive.

- Ablation methodology. (a) *Grading asymmetry* — two of six ablations (`perf-A` and the C2 sequel) published a rubric hash *before* results; the other four are honestly-run but post-hoc graded, so retrofitting is disavowed, not detectable. `perf-A` is the *only* ablation whose both arms ran under a pre-published rubric; C2's pre-registration covers its rubric and its fresh (Arm B) runs, but its Arm A was scored post-hoc from record ((c) below), so C2 sits partly in both categories. (b) *Isolation is a claim, not a given* — the harness pre-injects a fresh agent's cwd, git status, and memory *index*, so "blind" arms are mitigated, not sterile; the cleanest arms used fictional subjects (emu-B) or audited each agent's actual tool calls. (c) *Historical arms are not re-runnable* — some context-carrying arms (the INT8 ablation's Arm A and C2's Arm A) are transcripts of past work (the answer already known), so they bound rather than measure. (d) *N is small* (1–8/arm) and outcomes are often categorical (no variance). Each is stated in the individual `ablation_*.md`.
- $N{=}1$, one operator, one workstation, one model family.
- Self-authorship, including *this reframe* — written by an interested lead who is also the author of the substrate under evaluation (§V-C addendum), corrected by peers. We prefer machine evidence throughout; the RECALLED tier is defined in Table II and *is applied nowhere in this paper*, which is the honest report — an earlier draft promised the tag and used it once, on a claim now withdrawn. A provenance tier that is never applied is a tier the reader should not be asked to trust in.
- The bystander-catch corpus contracted under audit. While preparing this paper, two claimed cross-session catches did not survive record-checking — a "`api-svc` tag-flip caught by a peer" was in fact self-caught, and "slam-A caught four independent errors" had no case-file or bus record — and both were removed (§V-C). We report the contraction rather than quietly correcting it, because it is itself evidence for the provenance discipline the paper claims: the corpus is what the record supports and what its own rubric admits (13 attested, 12 classifiable), not what narration remembered.

VIII. Future Work: A Pre-Registered Replication

The paper's central threat is $N{=}1$ — one domain (systems/tooling), one operator, one workstation, one model family. We therefore *pre-register* the replication that would falsify or corroborate the claims, and state its prediction before running it (predict-then-run, a discipline the fleet itself enforces):

- Vary the confounding axes, not the surface. Repeat the experiment on a structurally different domain — ideally not software (e.g. scientific literature synthesis, hardware/product design, operations, or creative work), where "a task" and "expertise" mean something different in kind. Ideally with a different human architect, to remove the "maybe it is this operator" confound.
- Hold the *method* constant, vary the *content*. The claim under test is not "the tool runs on another codebase" but that *the composition* — persistent context-heavy peers + a shared bus + verified orders + a human gating irreversibility — reproduces the signatures this deployment did establish — peer review catching the lead's blind spot, and a substrate the fleet designed itself — and *tests for the first time* the ones it could not: compounding competence (a downward per-task-cost trajectory, via a protocol that manipulates session context rather than a prompt paragraph), and whether accumulated context confers an advantage in task origination, the step our ablations held fixed.
- Prediction (falsifiable): if the principles are real and not artifacts of this domain/operator, the *same* signatures appear on the new "thing"; if they are artifacts, they do not. A null result is as informative as a positive one and will be reported as such.
- Feasibility: the replication needs no new system — only a new *goal* pointed at a fleet (`project new <thing>`), which is precisely what makes the method's claim to generality testable rather than rhetorical.

This is the sequel experiment; naming it here is the honest acknowledgement that a single deployment, however rich, establishes a *method worth replicating*, not a law.

IX. Conclusion

Conductor is an experience report on a nameable composition — one model, durable per-session memory, a shared bus, a human gating only irreversibility — that co-designed and hardened its own coordination substrate in vivo. Its headline intuition — that accumulated context makes a session more capable — we could not test. An earlier version reported that it does not hold; we withdraw that, because the ablation behind it never varied the construct it named and its falsifier could never fire. The corrected experiment is pre-registered and witnessed and *cannot be run on this deployment at all*: a fleet with one session per repository has no second session carrying the same history.

What the deployment does establish is smaller and sturdier. Silent coordination defects fall into a small number of recurring shapes, and each names a place a verification must go. Peers with different vantages catch what an author and an operator miss — including, repeatedly, in this paper: three of the claims above were withdrawn on defects found by other sessions and an outside reviewer after we thought the work was finished, and every one of those defects ran in the direction that flattered us. The process that produced those corrections — peers refuting the lead on the record, an ablation refuting its own author — is the mechanism under study, and it is the part we would carry to another deployment.

The reusable contribution is the method: in-vivo adversarial co-design by persistent, context-divergent peers under a reversibility-gating human, offered as an existence proof with a stated mechanism, not a validated superiority. Its cleanest demonstration is this paper's own review. The internal panel,

at the deployment vantage, caught substance — a circularity, a single-model confound, unrun ablations; independent external review, at the submission vantage, caught what the panel structurally could not — cross-section consistency, arithmetic, and venue compliance. Which vantage caught which defect class is checkable in the archived draft series and the paper's own git history: the vantage bound of §V-C, one level up.

Three secondary observations we offer for trial beyond this deployment: the silent-defect taxonomy and its TAG→ABLATE→HARVEST remedy split (§V-A); the non-independence discipline — before crediting two results as corroboration, check for a shared generative mechanism, and discount if found (§V-C); and the durability ladder — untracked → committed → pushed — because a claim a reviewer cannot resolve is, at that moment, not yet evidence. The working summary for practitioners is §VI: fund the carrier, use sessions for speed and routing, audit for the regression, copy the cheap primitives.

Appendix: Evidence vs. Illustration

What the replication package can and cannot resolve. We state this plainly because the paper sets the bar itself (§V): *a claim a reviewer cannot resolve is, at that moment, not yet evidence*. RESOLVABLE: the substrate repository (git history, the `ablation_*.md` files, the pre-registered specs and their hashes, `history.jsonl`-derived series). **NOT RESOLVABLE: the commits backing the RQ3 cross-session catch table.** Those defects were fixed in separate hardware- and vendor-specific repositories that are not ours to release, so their hashes do not resolve against the released package. By our own rule the RQ3 corpus is therefore *attested but not reviewer-checkable*, and we mark it so rather than let a pinned-looking hash imply otherwise. **Evidence (MEASURED):** the mechanical record (git, bus, `history.jsonl`, spend meter), the six A/B ablations (the `ablation_*.md` files, replication package), and the pre-registered RQ5 baseline (`cases_rq5-baseline.md` plus the frozen spec/test and both arms' diffs). These carry the headline claims. Illustration (dataset, not evidence): twenty first-person `cases_*.md` specimens (from the 19 case-writing sessions of §V-F — one session contributed two) + the platform specimen `cases_cleanup-timer`, retained to illustrate design patterns, not cited as proof of any headline claim. Curated to five in-body illustrations, each mapping to a distinct claim:

1) `perf-A` — paper-as-instrument (§V-F): a 46-day latent defect surfaced *by writing the case*, and an ablation that *refuted the author's own Case 2* — the reflexive mechanism at its sharpest.
2) `net-emu` ↔ `bench-A` — RQ3 bystander/vantage: observer-vs-subject mutual correction (bench-A's "+64 s stall" was its own backlog; net-emu's guest-side clock settled it) — each catches what the other structurally cannot.
3) `mcu-emu` (ablation) — the §V-A class-5 *thoroughness-regression* specimen (context pattern-matched a staged answer and stopped; the blind arm caught the second register class).
4) `app-A` (a consumer app handling private user data) — a 5th independent derivation of "the model narrates, deterministic code owns the value" reached *from privacy* — convergence that survives a change of *force*.
5) `image-gen` — lived-expertise routing (§VI): the GPU-lease-lying / cost-blow-up incidents it was assigned to document *because it lived them* — the motivating case for routing on history.

Several cases were corrected during the paper's own writing (perf-A's Case 1 severity downgraded from user-visible to latent; perf-B's propagated overstatement retracted), logged as §V-F specimens.

Disclosure: Use of Generative AI

This paper reports on, and was substantially produced by, a fleet of *Claude Code* (Anthropic) sessions — the same system the paper studies — under the human author's direction. We disclose the division of labour precisely, because for this subject it is itself a finding. The agent sessions: drafted the prose of every section across the full draft series; *adversarially peer-reviewed* one another's drafts on the shared bus and produced the three-lens review that forced the v2 and v4 reframes; designed and executed the six A/B ablations — two of them (`perf-A` and the C2 open-search-ceiling sequel) pre-registered — and the RQ5 baseline; and mined the git, bus, and `history.jsonl` records for the quantitative evidence. The human author set the goal, made every irreversible decision (each `git push` was human-gated), chose the framing and the venue, took final responsibility for every claim, and is the sole party accountable for this submission. No text was accepted without human review. All quantitative results were re-derived from the primary records cited. Independent human external review (multiple rounds) then caught defect classes the fleet's own review structurally could not — cross-section consistency, arithmetic, and venue-compliance — and drove the final revisions; §IX notes why this split falls out of the paper's own thesis. Per ACM/IEEE authorship policy, the generative-AI system is *not* an author; its contribution is disclosed here and acknowledged below.

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